EE3032 Introduction to VLSI—Midterm

1. (30 points) Answer the following problems.
   - (a) Explain the purpose of design rules in VLSI design. (4 points)
   - (b) Realize the Boolean function $f = \overline{x_1 + x_2 x_3 + x_4}$ with static CMOS logic. (4 points)
   - (c) What is antenna effect? (3 points)
   - (d) List four properties of static CMOS logic. (4 points)
   - (e) Explain why adding a guard ring can reduce the latchup effect. (4 points)
   - (f) List two possible approaches for alleviating the effect of crosstalk on two neighboring parallel interconnects. (4 points)
   - (g) Explain the energy consumption of an inverter when the inverter performs a switching. (4 points)
   - (h) List the three major components of the load capacitance of a CMOS logic gate in a CMOS circuit. (3 points)
2. (10 points) Figure 1 depicts the process cross section of a CMOS inverter.
   - (a) Draw the circuit level model for modeling the latchup phenomenon. (5 points)
   - (b) Explain the phenomenon of latchup based on the model derived in (a). (5 points)

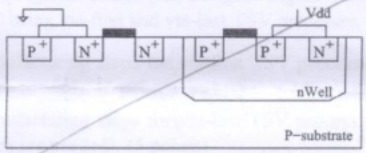


Figure 1: Process cross section view of a CMOS inverter.

3. (10 points) Figure 2 depicts two designs of inverter chains. Assume that the equivalent resistance of a minimum size W/L NMOS is R and $\mu_n = 2\mu_p$. The gate capacitance of a W/L transistor is C_g. Also, the parasitic capacitance of drain/source and wires are neglected. Which inverter chain consumes lower short-circuit power. Explain your answer.

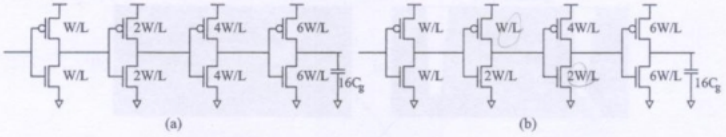


Figure 2: Inverter chains.

4. (30 points) Figure 3(a) depicts a CMOS logic circuit. Assume that the delay of an inverter is 1 delay unit, where the inverter size is $W_P/L_p = 2\mu m/0.18\mu m$ and $W_n/L_n = 1\mu m/0.18\mu m$ and the input capacitance of the inverter is C_i. Also, assume that the size of each NMOS transistor and PMOS transistor of all the other gates is the same as that of the inverter. When the CMOS logic circuit is operated, possible functional inputs of the circuit are {01001, 11011, 11100, 01101}. Also, these inputs are applied in sequence and the frequency is f and the power supply is V_{dd}.

(a) Estimate the peak power of this circuit in terms of f, C_i, and V_{dd}. (10 points)

(b) Determine the delay of each gate in terms of delay unit (neglect the diffusion and wire capacitance). (10 points)

(c) Assume that a dual-V_t process can be used to implement the CMOS circuit. Draw the transistor schematic diagram of the CMOS circuit in terms of the symbols shown in Fig. 3(b) such that the leakage power of the CMOS circuit is minimal without sacrificing the speed of the CMOS circuit. (10 points)

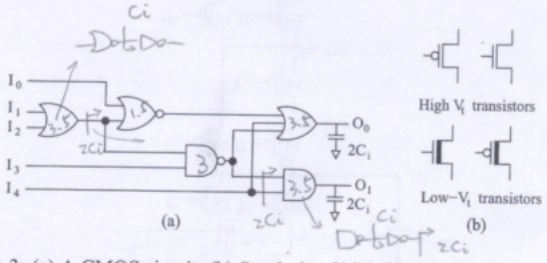


Figure 3: (a) A CMOS circuit. (b) Symbols of high-V_t and low-V_t transistors.

5. (10 points) Three-dimensional (3D) integration technology using through silicon via (TSV) is an emerging technology for integrated circuit (IC) designs. Figure 4 shows partial manufacturing steps for implementing a 3D IC using via-first and via-last TSV processes.

(a) List the correct manufacturing steps for via-first TSV process using the alphabet in sequence (hint: only 3 steps are needed). (5 points)

(b) List the correct manufacturing steps for via-last TSV process using the alphabet in sequence (hint: only three steps are needed). (5 points)

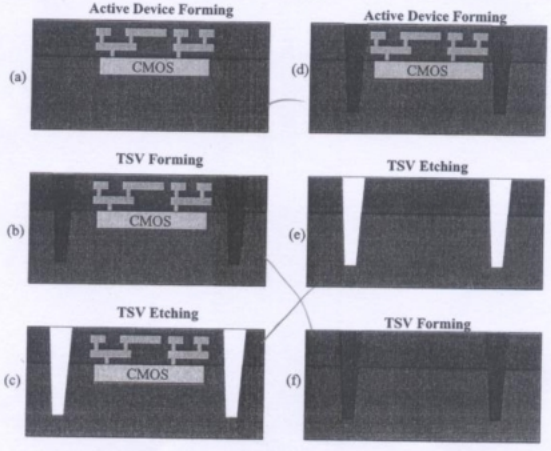
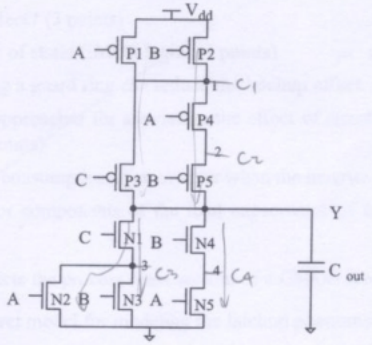


Figure 4: Partial manufacturing steps for implementing a 3D IC using via-first and via-last TSV processes.

6. (15 points) The figure shown below depicts a transistor-level schematic of the Boolean function Y.

(a) Show the corresponding gate-level schematic of the Boolean function Y. (5 points)

(b) Assume that the resistances of the transistors P1, P2, P3, P4, P5, N1, N2, N3, N4, and N5 are R_{P1}, R_{P2}, R_{P3}, R_{P4}, R_{P5}, R_{N1}, R_{N2}, R_{N3}, R_{N4}, and R_{N5}, respectively. Also assume the parasitic capacitance of nodes 1, 2, 3, and 4 are C_1, C_2, C_3, and C_4, respectively. Calculate the worse-case rise time and fall time of the circuit with simple RC delay model. (10 points)



AB + C(A+B)